NCC CADET MANAGEMENT SYSTEM
SOFTWARE TEST CASE REPORT

Document Title	Software Test Case Report — NCC Cadet Management System
Project Name	NCC Cadet Management System
Version	1.0.0 (Final Release)
Prepared By	Development Team
Date of Testing	April 2026
Testing Type	Functional, Integration, Security, Boundary, UAT
Testing Tool	Manual (Postman for API), Browser-based E2E
Total Test Cases	40
Passed	40 (100%)
Failed	0
Application URL	https://ncc-1-maharashtra.vercel.app/

1. TEST SCOPE & OBJECTIVES

This document records all test cases executed for the NCC Cadet Management System covering functional, integration, security, and boundary testing across all six application modules. Testing was performed against both the Officer and Cadet user roles to validate RBAC enforcement, data integrity, and UI correctness.

SCOPE ITEM	DETAIL
Modules Tested	Authentication, Cadet Profile, Camp Management, Attendance, Achievements, Study Materials
Roles Tested	Administrative Officer, Cadet
Test Environments	Local (Docker), Production (Vercel + Supabase)
Out of Scope	Mobile responsiveness, load/stress testing, third-party SMS/email integrations (future roadmap)

2. MODULE 1 — AUTHENTICATION & AUTHORIZATION

Test Cases: TC-001 to TC-008

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-001	**Auth**	Valid Officer Login	Functional	Officer account exists in DB	1. Open login page 2. Enter valid officer email & password 3. Click Login	Email: officer@ncc.in Pass: Test@1234	Redirect to Officer Dashboard with KPI panel	Redirected to Officer Dashboard	✓ **PASS**
TC-002	**Auth**	Valid Cadet Login	Functional	Cadet account exists in DB	1. Open login page 2. Enter valid cadet credentials 3. Click Login	Email: cadet@ncc.in Pass: Cadet@123	Redirect to Cadet Portal screen	Redirected to Cadet Portal	✓ **PASS**
TC-003	**Auth**	Invalid Password Login	Negative	User account exists	1. Enter correct email 2. Enter wrong password 3. Click Login	Email: officer@ncc.in Pass: wrongpass	Error: 'Invalid credentials' shown, no login	Error message displayed, login denied	✓ **PASS**
TC-004	**Auth**	Non-existent User Login	Negative	No pre-condition	1. Enter unregistered email 2. Enter any password 3. Click Login	Email: ghost@xyz.com Pass: abc123	Error: 'User not found', stay on login page	User not found error shown	✓ **PASS**
TC-005	**Auth**	Cadet Registration	Functional	No duplicate enrollment	1. Open register page 2. Fill all fields 3. Submit form	Name: Arjun Patil Enroll: MH/2024/A/0042 Wing: Army	Account created, redirected to cadet login	Account created successfully	✓ **PASS**
TC-006	**Auth**	Duplicate Enrollment Block	Boundary	Enrollment MH/2024/A/0042 exists	1. Register with existing enrollment number	Enroll: MH/2024/A/0042	409 Conflict — 'Enrollment already registered'	Duplicate blocked, 409 returned	✓ **PASS**
TC-007	**Auth**	Invalid Enrollment Format	Validation	None	1. Enter enrollment in wrong format 2. Submit	Enroll: 12345ABC	RegEx fails, inline error: 'Invalid format [State]/[Year]/[Wing]/[Num]'	Validation error displayed	✓ **PASS**
TC-008	**Auth**	JWT Token Expiry	Security	User is logged in	1. Login 2. Wait for token to expire (or manually expire) 3. Attempt API call	Expired JWT token	401 Unauthorized — session expired message	Session expired, redirected to login	✓ **PASS**

3. MODULE 2 — CADET PROFILE MANAGEMENT

Test Cases: TC-009 to TC-014

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-009	**Cadet Profile**	View Own Profile	Functional	Cadet is logged in	1. Login as cadet 2. Navigate to My Profile	Cadet JWT token	Display: name, rank, wing, enrollment number	Profile displayed correctly	✓ PASS
TC-010	**Cadet Profile**	Officer Views Cadet List	Functional	Officer is logged in	1. Login as officer 2. Navigate to Cadet Management	Officer JWT token	Full list of registered cadets with details	All cadets listed	✓ PASS
TC-011	**Cadet Profile**	Officer Updates Cadet Rank	Functional	Cadet TC-005 exists	1. Officer opens cadet record 2. Changes rank to 'Lance Corporal' 3. Saves	Cadet ID, New rank: Lance Corporal	Rank updated in DB, confirmation message	Rank updated successfully	✓ PASS
TC-012	**Cadet Profile**	Cadet Cannot Edit Own Profile	Security	Cadet is logged in	1. Cadet attempts PUT /cadets/:id via console/Postman	Cadet JWT, modified rank field	403 Forbidden — cadet cannot modify records	403 returned, no changes made	✓ PASS
TC-013	**Cadet Profile**	Cadet Cannot View Other Cadets	Security	Two cadet accounts exist	1. Login as Cadet A 2. Attempt GET /cadets (all-cadets endpoint)	Cadet A JWT token	403 Forbidden — role insufficient	403 Forbidden returned	✓ PASS
TC-014	**Cadet Profile**	Empty Cadet List Handling	Boundary	No cadets registered	1. Login as officer 2. Open cadet list	Officer JWT token	'No cadets registered yet' message shown	Empty state message displayed	✓ PASS

4. MODULE 3 — CAMP MANAGEMENT

Test Cases: TC-015 to TC-022

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-015	**Camps**	Officer Creates New Camp	Functional	Officer is logged in	1. Navigate to Camp Management 2. Click Add Camp 3. Fill all fields 4. Submit	Name: RDC 2024 Location: Pune Start: 10-Dec-2024 End: 20-Dec-2024	Camp created, appears in camp list	Camp created and listed	✓ PASS
TC-016	**Camps**	Cadet Views Available Camps	Functional	At least 1 camp exists	1. Login as cadet 2. Navigate to Camps section	Cadet JWT token	List of all upcoming camps with details	Camps displayed correctly	✓ PASS

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-017	**Camps**	Cadet Applies for Camp	Functional	Camp TC-015 exists; cadet logged in	1. Click Apply on RDC 2024 2. Confirm application	Cadet ID + Camp ID	Status changes to 'Applied'; enrollment record created	Application recorded successfully	✓ PASS
TC-018	**Camps**	Duplicate Camp Application	Boundary	Cadet already applied for TC-015	1. Cadet attempts to apply again for same camp	Cadet ID + Camp ID (existing)	Error: 'Already applied for this camp'	Duplicate blocked, error shown	✓ PASS
TC-019	**Camps**	Officer Views Applicant Manifest	Functional	Cadet TC-017 applied	1. Officer opens RDC 2024 2. Views applicants tab	Officer JWT token	List of all cadets who applied with their details	Manifest displayed correctly	✓ PASS
TC-020	**Camps**	Officer Edits Camp Details	Functional	Camp TC-015 exists	1. Officer edits camp end date 2. Saves changes	New End Date: 22-Dec-2024	Updated date saved, reflected in all views	Update saved correctly	✓ PASS
TC-021	**Camps**	Cadet Cannot Create Camp	Security	Cadet is logged in	1. Cadet attempts POST /camps via Postman	Cadet JWT, camp data	403 Forbidden	403 returned	✓ PASS
TC-022	**Camps**	Camp with Past Date Validation	Validation	Officer is logged in	1. Attempt to create camp with start date in the past	Start: 01-Jan-2020	Validation error: 'Start date cannot be in the past'	Validation error shown	✓ PASS

5. MODULE 4 — ATTENDANCE MANAGEMENT

Test Cases: TC-023 to TC-028

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-023	**Attendance**	Officer Marks Attendance	Functional	Officer logged in; cadets exist	1. Open Attendance module 2. Select today's date 3. Mark Present/Absent/Excused for each cadet 4. Submit	Date: today Cadet statuses	Attendance records saved for all cadets	Records saved to DB correctly	✓ PASS
TC-024	**Attendance**	Cadet Views Own Attendance	Functional	TC-023 completed	1. Cadet logs in 2. Views attendance history	Cadet JWT token	Chronological attendance log with date and status	Attendance history displayed	✓ PASS
TC-025	**Attendance**	Duplicate Attendance Block	Boundary	Attendance for today already exists	1. Officer tries to mark attendance again for same date	Same date + cadet IDs	Error: 'Attendance already marked for this date'	Duplicate blocked	✓ PASS

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-026	**Attendance**	Attendance Status Enum Validation	Validation	Officer logged in	1. Attempt to submit invalid status via API	Status: 'Holiday' (invalid)	422 Unprocessable Entity — status must be Present/Absent/Excused	422 returned	✓ PASS
TC-027	**Attendance**	Cadet Cannot Mark Attendance	Security	Cadet is logged in	1. Cadet attempts POST /attendance via Postman	Cadet JWT, attendance data	403 Forbidden	403 returned	✓ PASS
TC-028	**Attendance**	Empty Date Attendance View	Boundary	No attendance on selected date	1. Officer selects a date with no records	Date with no data	'No attendance records for this date' message	Empty state handled	✓ PASS

6. MODULE 5 — ACHIEVEMENTS

Test Cases: TC-029 to TC-034

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-029	**Achievements**	Cadet Submits Achievement	Functional	Cadet is logged in	1. Open Achievements 2. Click Submit 3. Fill title, date, description 4. Submit	Title: NCC 'A' Certificate Date: 15-Mar-2024	Achievement saved with status=Pending	Saved, status shows Pending	✓ PASS
TC-030	**Achievements**	Officer Views Pending Achievements	Functional	TC-029 completed	1. Officer opens Achievements panel 2. Views pending list	Officer JWT token	List of all Pending submissions with cadet details	Pending list displayed	✓ PASS
TC-031	**Achievements**	Officer Approves Achievement	Functional	TC-030 visible	1. Officer clicks Approve on TC-029 2. Confirms	Officer JWT + Achievement ID	Status changes to Approved; cadet portfolio updated	Status updated to Approved	✓ PASS
TC-032	**Achievements**	Officer Rejects Achievement	Functional	New pending achievement exists	1. Officer clicks Reject 2. Adds rejection note	Officer JWT + Achievement ID	Status changes to Rejected; reason stored	Status updated to Rejected	✓ PASS
TC-033	**Achievements**	Cadet Cannot Approve Own Achievement	Security	Cadet has a Pending achievement	1. Cadet attempts PATCH /achievements/:id via Postman	Cadet JWT, status=Approved	403 Forbidden	403 returned	✓ PASS
TC-034	**Achievements**	Empty Title Validation	Validation	Cadet is logged in	1. Submit achievement with empty title field	Title: (empty)	Inline error: 'Title is required'	Validation error shown	✓ PASS

7. MODULE 6 — STUDY MATERIALS

Test Cases: TC-035 to TC-038

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-035	**Study Materials**	Officer Uploads Material	Functional	Officer is logged in	1. Open Study Materials 2. Click Upload 3. Add title, description, file URL 4. Submit	Title: NCC Drill Manual URL: supabase-storage-url	Material appears in repository list	Material uploaded and listed	✓ PASS
TC-036	**Study Materials**	Cadet Views & Downloads Material	Functional	TC-035 completed	1. Cadet opens Study Materials 2. Clicks on uploaded PDF	Cadet JWT token	PDF opens/downloads via secure URL	File accessed successfully	✓ PASS
TC-037	**Study Materials**	Cadet Cannot Upload Material	Security	Cadet is logged in	1. Cadet attempts POST /materials via Postman	Cadet JWT, material data	403 Forbidden	403 returned	✓ PASS
TC-038	**Study Materials**	Missing File URL Validation	Validation	Officer is logged in	1. Submit material without file_url	file_url: (empty)	Validation error: 'File URL is required'	Validation error shown	✓ PASS

8. SECURITY TEST CASES

Test Cases: TC-039 to TC-040

TC-ID	MODULE	TEST CASE	TYPE	PRE-CONDITIONS	TEST STEPS	INPUT DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-039	**Security**	SQL Injection Attempt	Security	None	1. Enter SQL in login email field 2. Submit	Email: ' OR 1=1 -- Pass: anything	Login rejected; no DB error exposed; 401 returned	401 returned, no DB error	✓ PASS
TC-040	**Security**	Access Admin API Without Token	Security	None	1. Call GET /cadets with no Authorization header	No token	401 Unauthorized	401 returned	✓ PASS

9. TEST EXECUTION SUMMARY

MODULE	TOTAL TCs	PASSED	RESULT
Authentication & Authorization	8	**8**	All Passed ✓

Cadet Profile Management	6	**6**	All Passed ✓
Camp Management	8	**8**	All Passed ✓
Attendance Management	6	**6**	All Passed ✓
Achievements	6	**6**	All Passed ✓
Study Materials	4	**4**	All Passed ✓
Security	2	**2**	All Passed ✓
TOTAL	**40**	**40**	**100% Pass Rate — APPROVED FOR PRODUCTION ✓**